Modeling Honolulu Airport Hub Operations Under Reduced Air Traffic Capacity Constraints

Abstract

Hawaiian Airlines provides domestic and international flights to its patrons within the Hawaiian islands and beyond. Developing plans and methods for efficient fleet assignment is critical for maintaining operations, ensuring passenger safety and meeting flight demand. To achieve a concise fleet assignment schedule, airlines coordinate many factors at their major hubs, such as flight demand, airspace utilization, fleet availability and variable costs associated with route operation, like crew compensation and fuel.

Disruptions to standard airport operations are frequent and increasingly sustained, forcing airlines to adjust operations at their airports, especially at major hubs. Shortages of federal air traffic control and lapses in federal funding in particular reduce hourly flight traffic, driving airlines like Hawaiian Airlines to develop contingency scheduling to meet passenger demand, potentially incurring greater operating costs as air travel becomes increasingly unpredictable. This study includes analyses of round-trip flights from the Hawaiian Airlines hub in Honolulu to other Hawaiian islands and the costs associated with their operations. A mixed-integer linear programming approach is used to determine the optimal scheduling in an extended operations window with elevated operations costs. The model demonstrates how Hawaiian Airlines can ensure sufficient service from Honolulu to island destinations despite reduced air traffic flow at a minimum operating cost.

Keywords

aircraft fleet assignment, mixed-integer linear programming, cost minimization, capacity constraints

Introduction

Hawaiian Airlines is a subsidiary of Alaska Airlines that was recently acquired in 2025. The airline has a history that dates back to January 30, 1929, when it was founded under Inter-Island Airways (Bits of Aloha 2025). It consists of hubs in the Daniel K. Inouye International Airport (HNL) in Honolulu and the Kahului Airport (OGG) in Maui. The airline serves flights to other airports in the Hawaiian islands, along with airports across the mainland United States from coast to coast, including Seattle to New York. Along with domestic flights, Hawaiian Airlines also serves international flights primarily to Japan, Korea, Canada, and Australia (Hawaiian Airlines 2026b).

The HNL hub accounts for a substantial portion of those total flights, serving as the primary connection base for most inter-island travel throughout Hawaii. Bureau of Transportation Statistics air-travel records on February 26, 2025, for example, showed 83 Hawaiian Airlines departures and 82 Hawaiian Airlines arrivals at the HNL hub. Inter-island flights made up 64 of those departures and 64 of those arrivals. Inter-island flights departed within windows of approximately 5:00 AM to 8:00 PM, providing service between Honolulu and Kahului, Kailua/Kona, Lihue, and Hilo (Bureau of Transportation Statistics 2025a, 2025b). Meeting passenger demand for inter-island travel at the HNL hub is critical for Hawaiian Airlines' long-term stability as the primary airline operating in the region. Disruptions to operations can damage brand perception and customer loyalty, airline profits long-term, and allow larger airline competitors to enter the marketplace and potentially push Hawaiian Airlines out of the market.

Developing plans to sustain inter-island operations during times of uncertainty or constrained air travel at the HNL hub is critical. Certain disruptions to operations, like weather events, are acute and likely would not damage Hawaiian Airlines’ operations in the region

long-term. However, with growing shortages of air traffic control and essential airline personnel, coupled with long-lasting government shutdowns, airlines like Hawaiian Airlines must develop contingency plans to sustain operations during lengthy disruptions and protect their bottom line (United States Government Accountability Office 2025; Friedlander 2026).

This study addresses Hawaiian Airlines' problem of minimizing variable operations costs and meeting flight demand at the HNL airport hub during extended air traffic disruptions. Hawaiian Airlines' inter-island flight schedule primarily operates within a 5:00 AM to 8:00 PM departure window (Bureau of Transportation Statistics 2025a, 2025b). Outside of this window, there are potential costs associated with flight operations, like increased compensation for undesirable staffing slots. Airlines typically avoid flying routes with higher operations costs. Under normal conditions, they are able to meet demand without extending scheduling windows. However, with more frequent and sustained stresses on air traffic control systems, flight departures per hour are often reduced to ensure safety. For example, a federal government shutdown forced 10% air traffic reduction at over 40 U.S. airports in 2025, including HNL (Federal Aviation Administration, 2025). Reductions in hourly flight capacity out of the HNL hub could severely impact Hawaiian Airlines' ability to meet passenger demand within the typical operating window.

This study must determine how Hawaiian Airlines can meet daily flight demand under heightened traffic constraints at a minimized fuel and staffing cost. To do so, it proposes that utilizing early-morning and late-evening "grey-zone" departures during these long-term disruptions to ensure sufficient round-trip service between islands despite slightly higher operations costs. A mixed-integer linear programming model is developed to determine a

minimum cost schedule of round-trip flight operations from HNL to neighboring islands, satisfying daily route demand, aircraft availability, and HNL airport traffic capacity constraints.

Literature Review

Fleet Assignment Problem

The flight assignment problem (FAP) is a flight scheduling approach that optimizes revenue and minimizes costs while meeting flight route coverage requirements and fleet availability. Researchers and airlines have been using mathematical models to address this complex issue. The FAP is commonly modeled using optimization linear programming, but other approaches can be used such as simulation modeling. In recent years, these models have become more complex to account for real-world operations such as the maintenance of aircraft and flight demand. The models include the minimum amount of staff needed for each flight based on FAA regulations, different fleet types, limited number of aircraft and the maintenance of each aircraft. Airlines prepare fleet assignments over a year in advance since assignments are built on studies to determine consumer interest. Months before the flight takes off, crews are assigned to ensure qualified personnel and the minimum amount of personnel is aboard each aircraft while taking into consideration crew training and availability (Sciau et al. 2024). Airlines operate with slim profits and each assignment is carefully considered since any delays can lead to a loss in profit.

Fleet Assignment Problem: Integrating Real-World Constraints

Simpler FAPs often lack broader applicability, requiring additional optimization problem constraints that reflect airline services. To address this, Hane et al. devised a large-scale mixed integer program that utilizes a multicommodity flow network with constraints defined on a time expanded network (Hane et al. 1994). This model incorporated operational considerations in large-scale airlines modeling like destination node consolidation and elimination of missed

connections, building a flexible, realistic model applicable across a variety of regions and operational circumstances. The model's four major constraints ensured that each flight leg is flown by the same type of aircraft, that at each node an arrival flight triggers a departing flight, that linked flights use the same fleet, and that the number of aircraft cannot exceed the fleet size. These constraints reflect real-world flight considerations of fleet assignment problems. Hane et al.'s model has contributed to the development of advanced fleet models but its contributions are limited in their applicability since it assumes fixed time for each flight and a set number of passengers, addressed in novel iterations of the FAP (Birolini et al. 2021). Researchers have continued to propose new models that incorporate operational constraints like sufficient route coverage, fleet size, and runway duration. FAP has evolved since its inception to account for these parameters that substantially affect airline cost minimization efforts (Ahmed et al. 2024).

Linear programming is the most common method of approach for the fleet assignment problem all with maintenance and crew assignments. However, modeling approaches like the aviation integrated method (AIM) utilizes historical flight schedules and the size of the aircraft used in a multinomial logit model, weighting assignment probabilities based on past fleet scheduling (Bhadra 2005). This incorporates a realistic approach to actual application, favoring schedules that would cause less disruption in their initial implementation.

Further, innovations like the subnetwork fleet assignment model introduce tighter LP formulations, dividing national operations into regional subnetworks operating in tandem. This approach reduces LP complexity of large networks and integrates realistic operational strategies into proposed scheduling solutions (Barnhart, Farahat, and Lohatepanont 2009).

American and Delta Airlines: Applications of Fleet Assignment Optimization Strategies

Early application of fleet assignment models focused on real world applications. American Airlines and Delta Airlines developed different strategies pulling from these concepts at varying stages of their development. Abara utilized linear programming in practice for profit maximization along with cost minimization for departments at American Airlines to assess fleet planning and development of flight schedules. The model had binary, turnaround time constraints along with an aircraft balance that dictated that every arrival had to depart from the airport (Abara 1989).

Delta Airlines developed the Coldstart model, a daily repeated air schedule, with flexible air travel exception extensions. The model was built upon a previous model known as Warmstart that used fleet schedules made by schedulers and attempted to improve upon them using heuristics swaps (Subramanian et al. 1994). Delta developed this model to account for passenger spillage and uncertainty while being able to plan flight schedules without historical data. This model delegated flight routes as a network flight problem with nodes being the airport and the departure or arrival time. Along with the constraints that the assigned fleet will be able to accommodate the predicted number of passengers. The Delta Coldstart provided an early example of large-scale optimization implementation in airline operations and scheduling, serving as a strong conceptual reference for this analysis.

Optimizing Scheduling With Real-World Constraints in Hawaiian Airlines Hub-Operations

Prior research has focused on large-scale airline network optimization and fleet assignment models under typical operations. These models examine the complex scheduling challenges across entire airline systems. Building on base concepts of FAP, this analysis examines a regional subnetwork of Hawaiian Airlines operations at a localized hub under

heightened air traffic capacity constraints. Using airline historical data to determine typical airline operations, this model establishes daily route minimums without employing heuristic route assignments. This analysis integrates key real world constraints, like fleet availability and size limitations, in its approach. A mixed-integer linear program was developed to determine an optimal daily inter-island round-trip schedule at Hawaiian Airlines' Daniel K. Inouye International Airport (HNL) hub under reduced air traffic capacity. Incorporating air traffic capacity and extended hours of operations, this model builds on previous airline schedule modeling, examining how regional hubs can maintain service despite limited air traffic flow.

Methodology

Exploring Model Considerations

I. Time Slots and Flight Schedule Structure

This model considers the full 24-hour period, with hourly departure windows from 0:00 to 23:00. Each route from the hub HNL to an island destination is modeled as a round-trip flight, with a scheduled 2-hour window to complete the flight and turnaround time. This window accounts for the longest duration flight with an additional buffer for turnaround time (Bureau of Transportation Statistics 2025a, 2025b; Appendix A2.2, B.1).

II. Operational Costs

Fuel Costs:

To determine fuel costs, the fuel burn rate per hour was approximated for the aircraft servicing all inter-island fleets, the B717 regional jet (Hawaiian Airlines 2026a). The consumption rate per hour was converted into an estimated cost per hour using publicly available unit costs of jet fuel in early 2026. Route-specific total fuel costs were determined for each round-trip flight based on active flight time, reflecting the variation in each route's fuel usage

and subsequent cost (Appendix A.3; Bureau of Transportation Statistics 2025a, 2025b; Exxon Mobil Corporation 2022; EUROCONTROL 2024).

Peak-Hour Schedule Staffing Costs:

Flight crews for each round-trip flight must consist of 3 cabin crew and 2 pilots, as per federal regulations (Appendix A4.1-2; Code of Federal Regulations 2026a, 2026b). The hourly staffing costs are approximated using publicly available salary statistics for flight attendant (cabin crew) and pilot positions (Appendix A4.3; Bureau of Labor Statistics 2024a, 2024b). Total staffing wages for each round-trip flight are determined using the scheduled round-trip windows (2-hour) and calculated hourly wages (Appendix A4.3, A4.5).

Grey-zone Schedule Staffing Costs:

Higher wages associated with flight crews operating grey-zone scheduled departures are approximated using typical overtime pay standards as set by the Fair Labor Standards Act, 1.5 times the existing base hourly wage (Appendix A4.4.4; U.S. Department of Labor 2023).

III. Demand Assumptions

This model utilizes HNL inter-island Hawaiian Airlines flight operations from February 26, 2025 as documented by the Bureau of Transportation Statistics, as the basis for demand. This model assumes that demand has remained consistent between data collection and analysis dates. The most appropriate application of this model would be for a similar season of travel. Changes in individual passenger demand, ticket pricing, and route popularity fluctuations are considered outside the scope of this study. Demand is defined as the daily minimum round-trip flights occurring for each route, approximated from February 26, 2025, flight operations data (Appendix A1, Tables A.1-2; Bureau of Transportation Statistics 2025a, 2025b).

IV. Traffic and Operational Constraints

Peak-Hour Operations

This model identifies peak-hour operations for round-trip flights scheduled to depart HNL between 5:00 and 20:00 (Appendix B.1; Bureau of Transportation Statistics 2025a, 2025b).

Grey-zone Operations

This model identifies grey-zone operations for round-trip flights scheduled to depart HNL at 1:00, 2:00, 3:00, 4:00, 21:00 and 22:00. Flights departing at hours in the grey-zone are subject to higher staffing costs in this model (Appendix A4.4-5, B1, B3).

Blackout Zone

This model defines the blackout zone as the window of time in which no flights should be scheduled to depart, ensuring potential overnight fees are avoided for flight crews. These hours include 0:00 and 23:00. (Appendix B.4)

Normal Capacity Operations Scenarios

The approximated departure capacity during normal operations at HNL airport is approximated to be 4 departures per hour for this model (Appendix B.1, Bureau of Transportation Statistics 2025a, 2025b).

Reduced Capacity Operations Scenarios

During periods of disrupted operations, this model will reduce air traffic control capacity by approximately 25%, 10% to comply with historical federal ordinances during government shutdown induced staffing shortage and an additional 15% to allow buffer for unexpected aircraft events or emergencies (Appendix B.2, Federal Aviation Administration 2025). This reduces inter-island route departure capacity at HNL airport to approximately 3 departures per hour.

V. Additional Operations Considerations

Aircraft Capacity

Hawaiian Airlines' inter-island routes are entirely serviced by 20 Boeing B-717 regional aircraft (Hawaiian Airlines 2026a). Therefore, a maximum of 20 planes can be used to operate flights at a given time.

VI. Model Assumptions and Limitations

This model exhibits several limitations in its applicability to actual airline operations. This model assumes round-trip flights for all routes, ensuring all planes complete their routes back at the HNL hub. This simplifies the approach for tracking each plane within the 20-aircraft fleet for the scope of this study. Further, this model assumes all round-trip flight routes are accounted for in a 2-hour window, including flight-time and turnaround time. This model assumes no delays in those flight times or turnaround times to limit the scope of the analysis for this study.

Additionally, this model does not take into account individual passenger demand or fluctuations in demand and considers each flight operated to be fully-booked. Though important factors, this model approximates demand with required flights per route per day, aligning with Bureau of Transportation Statistics flight scheduling data pulled from February 26, 2025 (Bureau of Transportation Statistics 2025a, 2025b). Finally, this model does not take into account how ticket pricing impacts flight demand. This factor is outside the scope of the analysis for this model.

Defining the Model

This model is a mixed-integer linear programming model that uses binary outcomes for decision variables to decide if scheduling a given plane to depart HNL at a given time to fly a given route will minimize airline operations cost while meeting operations constraints.

Objective Function

Minimize costs represented by Z across all indices and subject to all constraints

$$\text{Min } Z = \sum_{r \in R} \sum_{t \in T} \sum_{p \in P} X_{r,t,p} * C_{r,t,p}$$

Sets and Indices

R = the set of all round-trip routes between HNL and other Hawaiian island airports

R includes Kahului, Kailua/Kona, Lihue, and Hilo round-trip routes out of the HNL hub

T = the set of all hourly time slots in the flight scheduling period, excluding the blackout zone, that a plane starts its round-trip service on route r

T = {1:00, 2:00, 3:00, …, 22:00}

P = the set of planes in the Hawaiian Airlines B717 fleet

P = {1, 2, …, 20}

Decision Variables

$X_{r,t,p}\{1\ if\ scheduled,\ 0\ otherwise\}$ where r = round-trip route, t = hourly time slot the route begins, p = plane operated

Parameters

F_r = required daily frequency of each round-trip route

X_{max} = maximum number of flight departures per hour on the ground at HNL, as per air traffic control safety limits

$C_{r,t,p}$ = costs (fuel, staffing base, staffing grey-zone wages) associated with each route flown, r, departing at time t on a given plane, p

$T_{greyzone}$ = time slots in which flights taking off at these times operate at higher costs {1:00, 2:00, 3:00, 4:00, 21:00, 22:00}

Constraints

Daily Route Flight Minimum Constraint

$$\sum_{t \in T} \sum_{p \in P} X_{r,t,p} = F_r \quad \text{for all r over R}$$

where r = round-trip route, t = hourly time slot route begins, p = plane operated

Round-Trip Plane Occupied Constraint

$$\sum_{r \in R} (X_{r,t,p} + X_{r,t+1,p}) \leq 1 \quad \text{for all planes in set P scheduled at time t}$$

This constraint limits the available scheduling time for each plane to 2 hours after it begins its round trip, the budgeted window for route completion across all routes.

Air Traffic Control Flight Departures per Hour Constraint

$$\sum_{r \in R} \sum_{p \in P} (X_{r,t,p}) \leq X_{max} \quad \text{for all t over T}$$

This constraint limits the number of departures per hour in slot time t.

This mixed-integer linear programming problem is solved using the Pulp optimization library in Python (Appendix D-F).

Results

The mixed-integer linear programming model for Hawaiian Airlines inter-island fleet assignment out of the HNL hub produced an optimal schedule that met airflow constraints, daily flight minimums and plane availability limits. To comply with reduced hourly air traffic flow and meet passenger flight demand, 16 flights were scheduled in the “grey-zone”, an extended scheduling window with slightly higher round-trip costs. These 16 flights serviced all destinations in the Hawaiian islands. Peak hours operations saw round-trip service to all destinations. All planes in the fleet were utilized in the daily schedule. The total minimized

variable costs of operating this fleet assignment schedule is approximately $182,950.15 (Appendix F). Sensitivity analysis was conducted by reducing the number of aircraft available. The model remained viable until the fleet was reduced to 6, which indicates the set parameters were too weak.

Discussion and Conclusion

Overall, this model provides a strong basis for building airline hub operations plans in response to restricted air traffic. This model prioritizes meeting passenger flight demand and minimizing variable costs associated with inter-island travel. Focusing on the primary hub of Hawaiian Airlines, the model is able to capture service provided to all surrounding local destinations. Further, though mainland and international destinations are not considered in this phase of the model, HNL serves as the primary port of exchange for travelers reaching island destinations throughout Hawaii. Ensuring consistent and frequent access to all islands, even in circumstances of restricted airflow, will allow Hawaiian Airlines to maintain its status as the primary air-carrier in the Hawaiian islands.

This model takes a simplified approach to hub operations, which limits its scope. Flights were scheduled to depart the HNL hub in 1 hour timeslots island-service routes are considered to fly a two-hour round-trip flight between the HNL hub and its destination. This does not fully reflect the potential variations in airtime, taxi-time and turnaround time for each destination, especially in the case of delays or rerouting. Further, this two-hour flighttime window limits the possibility of additional rest time in between outbound and return legs of each round-trip at the destination airport. Though a potential barrier in large-scale model adoption, this was an appropriate assumption for the purposes of Hawaiian Airlines' inter-island travel, which benefits from fast flight times and smaller passenger loads to allow for continuous service. In addition,

hourly binned departure scheduling limits scheduling times. This can be adjusted with shorter binned departure slots in broader adaptations of the model.

This model considered daily flight requirements as a proxy for passenger demand. Given that the daily flight requirements reflected that of Hawaiian Airlines' services on February 25, 2026, the demand for inter-island flights throughout the year or service at other airport hubs is likely not constant, despite the analysis treating it as such. The summer months and end of the year have the largest number of flights. February was one of the months with the lowest number of flights (Hawai'i Tourism Authority 2026). This model would be more comprehensive if it were able to incorporate long-term forecasting of demand and fluctuations in passenger travel trends to the Hawaiian islands. Applications of this model to other regional airline services would incorporate local fluctuations in passenger demand as well.

Conducting sensitivity analysis by changing the fleet size for the airlines, resulted in the model still being viable until the fleet size decreased to 6 planes and still be able to complete the scheduled flights. While mathematically the model works, realistically it could not be implemented by the airline for a variety of reasons. With the turnaround time for each aircraft designated at 2 hours, based on the model, aircrafts could make up to 11 flights in a single day. The planes would be constantly departing, thus leaving no time for crew change, refueling and cleaning the aircraft.

This model considered limited, variable costs associated with service along each route at each scheduled departure time. Fixed, largely consistent costs across all routes and departures, like catering costs or aircraft safety fees, were not considered. Fuel costs relied upon active flying times for each route. Inclusion of other factors contributing to fuel usage would be beneficial to developing a more robust representation of fuel cost per route flown. Further,

variations in crew compensation based on seniority and scope of responsibility were not considered, but could be relevant factors in accurately determining true crew compensation. For the purposes of this model, however, compensation did not vary based on route, as all crew members were compensated for the two-hour round-trip flight and turnaround time window. The airlines total fleet for inter-island was utilized for the making of this model such that it does not consider when airplanes are maintained and inspected, not every plane will be available to be utilized in a given day. The inclusion of grey-zone overtime staffing costs allowed the model to account for the increased costs of operating within off-peak hours. To make this model more comprehensive, additional cost penalties of operating within grey-zone could be applied, including potential costs of operating flights that are not completely full. For the purposes of this model, the grey-zone is a useful proxy for the potential increased operational costs airlines might incur as a result of air traffic limitations.

Overall, this analysis can serve as the foundation for the strategy of Hawaiian Airlines, and other air carriers, in preserving service despite disruptions or limitations in their typical operations schedules. Considering variable costs per flight and the daily flight demands, airline hubs will be equipped with the approaches and analyses necessary to adapt to a rapidly changing air travel industry.

References

Abara, Jeph. 1989. "Applying Integer Linear Programming to the Fleet Assignment Problem." *Interfaces*. 19 (4): 20-28. https://doi.org/10.1287/inte.19.4.20.

Ahmed, Alaa, Bassant Nabil, Hadi Fors, and Ahmad E. Elhabashy. 2024. "A Revised Formulation for the Airline Fleet Assignment Problem." In *Proceedings of the International Conference on Industrial Engineering and Operations Management*. Dubai, UAE, February 11-14, 2024. https://doi.org/10.46254/AN14.20240106.

Barnhart, Cynthia, Amr Farahat, and Manoj Lohatepanont. 2009. "Airline Fleet Assignment with Enhanced Revenue Modeling." *Operations Research*. 57: 231-244. https://doi.org/10.1287/opre.1070.0503.

Bhadra, Dipasis. 2005. "Choice of Aircraft Fleets in the U.S. Domestic Scheduled Air Transportation System: Findings from a Multinomial Logit Analysis." *Transportation Research Forum*. 44: 143-162. https://doi.org/10.22004/ag.econ.206775.

Birolini, Sebastian, Antonio Pais Antunes, Mattia Cattaneo, Paolo Malighetti, and Stefano Paleari. 2021. "Integrated Flight Scheduling and Fleet Assignment with Improved Supply-Demand Interactions." *Transportation Research Part B* 149: 162-180. https://doi.org/10.1016/j.trb.2021.05.001.

Bits of Aloha. 2025. "The History of Hawaiian Airlines & Its 2025 Alaska Merger." August 28, 2025. https://bitsofaloha.com/blogs/news/the-history-of-hawaiian-airlines-its-2025-alaska-merger.

Bureau of Transportation Statistics. 2026. "U.S. Airlines' January 2026 Aviation Fuel

Consumption Down 10.8%, and Fuel Cost per Gallon up 1.8% from December 2025." U.S. Department of Transportation. Accessed February 26, 2026. https://www.bts.gov/newsroom/us-airlines-january-2026-aviation-fuel-consumption-down-108-and-fuel-cost-gallon-18.

Bureau of Transportation Statistics. 2025a. "Airline On-Time Statistics: Detailed Statistics Departures, Daniel K. Inouye International (HNL), Hawaiian Airlines (HA)." U.S. Department of Transportation. https://www.transtats.bts.gov/ontime/departures.aspx.

Bureau of Transportation Statistics. 2025b. "Airline On-Time Statistics: Detailed Statistics Arrivals, Daniel K. Inouye International (HNL), Hawaiian Airlines (HA)." U.S. Department of Transportation. https://www.transtats.bts.gov/ontime/arrivals.aspx.

Code of Federal Regulations. 2026a. *Title 14: Aeronautics and Space*, §121.391, Flight Attendants. https://www.ecfr.gov/current/title-14/chapter-I/subchapter-G/part-121/subpart-M/section-121.391.

Code of Federal Regulations. 2026b. *Title 14: Aeronautics and Space*, §121.481, Flight time limitations: One or two pilot crews. https://www.ecfr.gov/current/title-14/chapter-I/subchapter-G/part-121/subpart-R/section-121.481.

EUROCONTROL. 2024. *EUROCONTROL Standard Inputs for Economic Analyses*. EUROCONTROL. https://www.eurocontrol.int/publication/eurocontrol-standard-inputs-economic-analyses

Exxon Mobil Corporation. 2022. "ExxonMobil Jet Fuel." ExxonMobil Commercial-Fuel. https://www.exxonmobil.com/en-bd/commercial-fuel/pds/gl-xx-jetfuel-series.

Federal Aviation Administration. 2025. “DOT & FAA Announce Temporary 10% Reduction in Flights at 40 Airports.” U.S. Department of Transportation, November 6, 2025. https://www.faa.gov/newsroom/dot-faa-announce-temporary-10-reduction-flights-40-airports.

Friedlander, Joshua P. 2026. "The Government Shutdown’s $6 Billion Toll on Travel and the U.S. Economy." U.S. Travel Association. Accessed March 10, 2026. https://www.ustravel.org/news/government-shutdowns-6-billion-toll-travel-and-us-economy.

Hane, Christopher A., Cynthia Barnhart, Ellis L. Johnson, Roy E. Martsen, George L. Nemhauser, and Gabriele Sigismondi. 1994. “The Fleet Assignment Problem: Solving a Large Scale Integer Problem.” *Mathematical Programming* 70: 211-232. https://doi.org/10.1007/BF01585938.

Hawaiian Airlines. 2026a. “Boeing 717-200.” Hawaiian Airlines. Accessed February 26, 2026. https://www.hawaiianairlines.com/content/our-services/at-the-airport/our-fleet/boeing-717-200?int=HA_WEB_OurFleet_borealis-grid3_CARD4%7C%7CBoeing%20717%20200%20Child%20Nav.

Hawaiian Airlines. 2026b. “Destinations.” Hawaiian Airlines. Accessed February 16, 2026. https://www.hawaiianairlines.com/en-au/content/destinations.

Hawai’i Tourism Authority. 2026. “Historical Visitor Statistics.” Hawaii Tourism Authority. Accessed March 10, 2026. https://www.hawaiitourismauthority.org/research/historical-visitor-statistics.

Sciau, Jean-Baptiste, Agathe Goyon, Alexandre Sarazin, Jeremy Bascans, Charls Prud’homme, and Xavier Lorca. 2024. “Using Constraint Programming to Address the Operational

Aircraft Line Maintenance Scheduling Problem." *Journal of Air Transport Management* 115:102537. https://doi.org/10.1016/j.jairtraman.2024.102537.

Subramanian, Radhika, Richard P. Scheff, John D. Quillinan, D. Steve Wiper, and Roy E. Marsten. 1994. "Coldstart: Fleet Assignment at Delta Airlines." *Interfaces* 24 (1): 104-120. https://doi.org/10.1287/inte.24.1.104.

U.S. Bureau of Labor Statistics. 2024a. "Airline and Commercial Pilots." *Occupational Outlook Handbook*. U.S. Department of Labor. Accessed March 10, 2026. https://www.bls.gov/ooh/transportation-and-material-moving/airline-and-commercial-pilots.htm.

U.S. Bureau of Labor Statistics. 2024b. "Flight Attendants." *Occupational Outlook Handbook*. U.S. Department of Labor. Accessed March 10, 2026. https://www.bls.gov/ooh/transportation-and-material-moving/flight-attendants.htm.

U.S. Department of Labor. 2023. "Handy Reference Guide to the Fair Labor Standards Act." Wage and Hour Division. Accessed March 10, 2026. https://www.dol.gov/agencies/whd/compliance-assistance/handy-reference-guide-flsa#1.

United States Government Accountability Office. 2025. "Air Traffic Control Workforce FAA Should Establish Goals and Better Assess Its Hiring Processes". https://files.gao.gov/reports/GAO-26-107320/index.html?_gl=1*1jmlewy*_ga*MTU2NjkwNDk5NC4xNzczMTc1ODI5*_ga_V393SNS3SR*czE3NzMxNzU4MjgkbzEkZzEkdDE3NzMxNzU4ODkkajYwJGwwJGgw.

Appendix

A: Building HNL Flights Dataset

A1: Approximating Passenger Demand

For this analysis, passenger demand is considered analogous to daily flight route demand. Flights considered in this analysis solely include inter-island routes to and from the HNL hub on February 26, 2025 (Bureau of Transportation Statistics 2025a, 2025b). Total flights operated on each inter-island route, including arrivals and departures, are detailed below:

Table A.1.1: Daily Inter-Island Flights, February 26, 2025*

Origin	Destination	Occurrences
Kahului	HNL	21
Kailua/Kona	HNL	15
Lihue	HNL	16
Hilo	HNL	12
HNL	Kahului	21
HNL	Kailua/Kona	15
HNL	Lihue	16
HNL	Hilo	12

For the purposes of this analysis, these arrival and departure route occurrences were used to approximate the demand for round-trip daily flights from HNL, as below:

Table A.1.2 Approximate Demand For Daily Round-Trip Inter-Island Flights*

Round-Trip Route	Approximate Daily Demand
HNL-Kahului-HNL	21
HNL-Kailua/Kona-HNL	15
HNL-Lihue-HNL	16
HNL-Hilo-HNL	12

*See AppendixData_HNL_arrivals_20250226.xlsx and AppendixData_HNL_departuress_20250226.xlsx for raw datasets and calculations.

A.2 Calculating Travel-Time

A.2.1 Calculating One Way Travel-Time

To approximate route travel time, the mean scheduled elapsed time of each one-way route was calculated across all inter-island routes, determining the one-way route travel time in minutes. This value was converted to hours.*

*See AppendixData_HNL_arrivals_20250226.xlsx and AppendixData_HNL_departuress_20250226.xlsx for raw datasets and calculations.

A.2.2. Calculating Round Trip Travel-Time

To determine round-trip route travel time, the mean travel time of individual legs of each round-trip route were added together. This value represents active travel time for each round-trip route in hours. All flight times are under 2 hours.**

**See AppendixData_hnl_flights_summary_20250226.xlsx for detailed calculations.

A3: Estimating Fuel Costs

A3.1: Fuel Cost Per Gallon

According to a Bureau of Transportation Statistics January 2026 Aviation Fuel report, the price of standard aircraft fuel is approximately $2.36 per gallon as of January, 2026. This metric is the most updated fuel pricing metric used to estimate costs associated with fuel consumption (Bureau of Transportation Statistics 2026).

A3.2: Fuel Cost Per Kg

To align with fuel consumption units, the cost per gallon of aircraft fuel is converted. The typical density of fuel is approximately 807.5 kg/m^3 at 15 degrees Celcius, according to Exxon Mobil Corporation. Given this density, the approximate cost of jet fuel is equivalent to 0.77 dollars/kg (see calculations below). Standard approximate conversions between liters and gallons are applied (Exxon Mobil Corporation 2022).

807.5 kg fuel	1 m^3	3.78 L	= 3.05 kg fuel
1 m^3	1000 L	1 gallon	1 gallon

2.36 dollars	1 gallon	= 0.77 dollars
1 gallon	3.05 kg	1 kg fuel

A3.3: Fuel Consumption Per Minute/Per Hour In-Flight

The Hawaiian Airlines inter-island fleet consists of Boeing B717 regional jets. Direct, publicly available fuel burn data per hour for these aircraft types is limited and highly variable. Manufacturer and regulatory agencies do not publish standardized fuel burn per hour rates due to these variations. For the purposes of this analysis, proxies for fuel consumption will be used, pulling proxies for fuel burn per hour data from the European Organization for the Safety of Air Navigation published Standard Inputs for Economic Analyses (EUROCONTROL 2024).

This model approximates B717 fuel consumption using the E190 regional jet, due to similar size, passenger capacity, and engine characteristics. In flight fuel consumption with 80% maximum useful load*** is approximated to be 31.2 kg/min. This was converted to 1872 kg/hr fuel consumed.**

***maximum useful load indicates the percentage of maximum allowable weight at takeoff. 80% metric was selected, given the assumption that flights will be operated completely full.

A3.4: Fuel Costs Per Flight

The fuel cost per hour of flight time 1441.44 kg/hr was calculated using hourly flight time fuel burn rates and fuel costs per kilogram. Utilizing total round-trip flight time (hours) for each route and fuel cost per flight hour, the fuel cost for each route was calculated.**

**See AppendixData_hnl_flights_summary_20250226.xlsx for detailed calculations.

<u>A4: Estimating Staffing Costs</u>

A4.1: Cabin Crew Minimum Staffing

United States Federal regulations require at least two flight attendants for all flights regardless of passenger load. For each additional 50 passenger unit, or portion of, an additional flight attendant is required (Code of Federal Regulations 2026a). B717 aircraft have a 128 passenger capacity and, therefore, require 3 flight attendants in the cabin crew.

A4.2: Pilots Minimum Staffing

All flights are staffed with a minimum of a two pilot crew to ensure appropriate rest time and availability of a pilot should an emergency occur. FAA regulations state that no pilot should

fly 8 hours without 24 hours rest. Given that no flight leg is longer than 16 hours, all routes are sufficiently staffed using two pilot crews (Code of Federal Regulations 2026b).

A4.3 Airline Staffing Wages: Peak Hours

Airline staff salaries are highly variable due to irregular scheduling, overtime pay, and long-haul flight bonuses. For the purposes of this analysis, cabin crew hourly wage is determined by taking the median annual salary of flight attendants and determining the hourly wage using a 40-hour work-week and 52 work-week year. The median annual pay of flight attendants in 2024 was $67,130, according to the Bureau of Labor Statistics (U.S. Bureau of Labor Statistics 2024a).

67130 dollars	1 year	1 work week	$ 32.27
year	52 work week	40 hours	hour

A similar approach is applied for determining hourly wage for pilots. 2024 median pilot annual salary was $198,100, according to the Bureau of Labor Statistics (Bureau of Labor Statistics 2024b).

198100 dollars	1 year	1 work week	$ 95.24
year	52 work week	40 hours	hour

A4.4 Airline Staffing Wages: Grey-zone Hours

For flights departing during grey-zone hours, the model assumes flight crew wages to be time and a half, reflective of standard overtime compensation as laid out by Fair Labor Standards Act (U.S. Bureau of Labor Statistics 2023).

A4.5 Total Staff Costs Per Flight

Aircraft staff are compensated for the entirety of the 2 hour round-trip flight time window for each route. Their hourly compensation varies based on peak hour or grey-zone departure times. For each route, the total number of crew members was multiplied by their respective hourly wage and the 2 hour scheduled working time. The total staff costs were determined for both grey-zone and peak hour operations.**

**See AppendixData_hnl_flights_summary_20250226.xlsx for detailed calculations.

A5 Flight Services Cost

Costs associated with in-flight services, like food and beverage services, and after-flight services, like cleaning, are assumed to be consistent across all inter-island B717 planes, which all carry the same passenger loads. These costs are fixed across all routes and reveal no additional information for comparing optimal flight schedules. Therefore, they are not included in the cost minimization objective.

B. Approximating Flight Operations

B.1 Peak-Hour Operations

This analysis focuses on flights operating on inter-island routes to and from the HNL hub on February 26, 2025 and their hours of operation. The schedule indicates inter-island flights are typically scheduled from 5:00 to 20:00, indicating a 15-hour window of peak-hour operations (Bureau of Transportation Statistics 2025a, 2025b). Other routes, including long-haul international flights, are scheduled outside of this window, but are outside the scope of this analysis.

During this peak-hour period, approximately 64 inter-island departures were scheduled (see Appendix A1 Table A1.1). This indicates a capacity of approximately 4.27 inter-island route departures per hour from HNL. It is unlikely that an airline would operate at the true maximum departure per hour capacity. Therefore, this standard operations capacity is rounded down to 4 inter-island route departures per hour. This provides additional buffer time so the existing peak-hour schedule can operate safely beneath the true limit.

B.2 Reduced Capacity Scenarios

In cases where flight operations are impacted by disruptions to air traffic control staffing or other air traffic safety changes, hourly departure capacity must be reduced to ensure passenger safety and effective aircraft monitoring. For example, during a 43-day federal government shutdown in 2025, aircraft operations were reduced by 10% across major airports, including HNL (Federal Aviation Administration 2025). To approximate the potential impact of these disruptions, departure flight capacity per hour will be reduced from 4 flights per hour to 3 flights per hour in reduced capacity modeling scenarios, indicating a 25% reduction in capacity. This 20% capacity allows for an additional 15% buffer to ensure departures maintain realistic buffer capacity in case of emergencies or unexpected air traffic events.

B.3 Grey-zone Operations

Grey-Zone operating hours refer to the 1:00, 2:00, 3:00, 4:00, 21:00 and 22:00 departure slots where round-trip flights can be scheduled at greater costs than those during peak-hours.

B.4 Blackout Zone

This refers to the hours of 0:00 and 23:00 for which no flights will be scheduled to depart. This allows additional buffer time to ensure planes are not continuously operated for over

24 hours. This also ensures that airline personnel are not staffed overnight to avoid additional staffing fees and potential overnight flight fees.

E. Performing the Mixed Integer Linear Programming for Fleet Assignment

To develop this mixed-integer linear program, data compiled from all previous appendix sections in 'island_flights_cleaned.csv' was used. All decision variables $x_{r,\,t,\,p}$ are included in the code. The corresponding costs for each decision variable are accounted for in the objective. Using the Pulp package, the mixed-integer linear program was solved, complying with the constraints laid out in previous sections. See attached file for full code.

island_flights_cleaned

roundtrip_destination	demand	flighttime	fuel_burned_hr	fuel_cost_hr	fuel_cost_flight	cabin	pilot	cost_staff_normal	cost_staff_grey
Kahului	21	1.36	1872	1441.44	1960.3584	3	2	669.82	861.87
Kona	15	1.59	1872	1441.44	2291.8896	3	2	669.82	861.87
Lihue	16	1.3	1872	1441.44	1873.872	3	2	669.82	861.87
Hilo	12	1.82	1872	1441.44	2623.4208	3	2	669.82	861.87

```python
# routes (defined by non-HNL destinations)
R = df["roundtrip_destination"].tolist()

# planes
P = list(range(1, 21))
    # planes id 1 through 20

# departure times
T = list(range(1, 23))
    # ONLY including valid departure times (none in the blackout zone)

# grey zone departure times
T_grey = [1, 2, 3, 4, 21, 22]
    # this is only needed to calculate cost coefficients

# demand dictionary for each route
F_r = {
    "Kahului" : 21,
    "Kona" : 15,
    "Lihue" : 16,
    "Hilo" : 12
}

# departure capacity
X_max = 3
```

```
# create decision variable set key
x_keys = list(itertools.product(R, T, P)) # this defines ALL the possible combinations of
    # route, time depart, and plane ID number in the entire parameter set
```

Define Cost Coefficients

```
cost_coeffs = {}

for r, t, p in x_keys:
    if t in T_grey:
        staffing_condition = "cost_staff_grey"
    else:
        staffing_condition = "cost_staff_normal"
    staff_cost = df.loc[df['roundtrip_destination'] == r, staffing_condition].iloc[0]
    fuel_cost = df.loc[df['roundtrip_destination'] == r, 'fuel_cost_flight'].iloc[0]
    total_cost = staff_cost + fuel_cost
    cost_coeffs[(r, t, p)] = total_cost
```

Define Decision Variables

```
# create decision variable set
x_vars = {}

for r, t, p in x_keys:
    key = (r, t, p)
    x_vars[key] = LpVariable(f"x_{r}_{t}_{p}", 0, 1, cat = "Binary")
    # this loops through all possible combinations of keyed decision variables, creates
    # a binary LP variable and then stores in x_vars with that key
```

Define Objective Function

```
# defines the problem
prob = LpProblem("problem", LpMinimize)
```

```
# defines the objective
prob += sum(cost_coeffs[k] * x_vars[k] for k in x_vars)
    # where k represents the key (r, t, p)
```

Define Constraints

Daily Route Flight Minimum Constraint

```
for r in R:
    prob += sum(x_vars[(r, t, p)] for p in P for t in T) == F_r[r]
```

Round-Trip Plane Occupied Constraint

```
for p in P:
    for t in T[:-1]: # for all takeoff times (2:00 to 21:00, but 22 avoided bc x23 does not exist
        prob += sum(x_vars[(r, t, p)] for r in R) + sum(x_vars[(r, t + 1, p)] for r in R) <= 1
```

Air Traffic Control Flight Departures per Hour Constraint

```
for t in T:
    prob += sum(x_vars[(r, t, p)] for r in R for p in P) <= X_max
```

Solve Problem

```
# solve the problem
status = prob.solve()
print(f"status={LpStatus[status]}")
```

```
Result - Optimal solution found

Objective value:                182950.15200000
Enumerated nodes:               0
Total iterations:               0
Time (CPU seconds):             0.02
Time (Wallclock seconds):       0.03

Option for printingOptions changed from normal to all
Total time (CPU seconds):       0.02   (Wallclock seconds):       0.05
```

F. Final Flight Schedule

AppendixF_HNL_schedule

Routes	Departure	PlaneID
Hilo	1	2
Hilo	1	7
Hilo	1	13
Kahului	2	5
Hilo	2	14
Lihue	2	15
Kona	3	8
Kona	3	9
Kahului	3	16
Lihue	4	7
Lihue	4	20
Kahului	5	2
Kona	5	6
Lihue	5	15
Kona	6	5
Kona	6	14
Hilo	6	20
Lihue	7	8
Lihue	7	10
Lihue	7	11
Kona	8	1
Kahului	8	9
Lihue	8	19
Kona	9	3
Kahului	9	12
Lihue	9	18
Hilo	10	10
Kona	10	17
Kahului	10	20
Kona	11	5
Kahului	11	8
Kona	11	19
Lihue	12	1
Kahului	12	9
Lihue	12	12
Kahului	13	3
Kona	13	15
Kahului	13	17
Kahului	14	6
Hilo	14	7
Lihue	14	20
Lihue	15	1
Kahului	15	9
Kona	15	13
Kona	16	6
Kahului	16	11
Kahului	16	18
Hilo	17	2
Kahului	17	4
Lihue	17	13
Hilo	18	3
Kahului	18	6
Kahului	18	9
Kona	19	11
Lihue	19	15
Lihue	19	20
Kona	20	4
Kahului	20	18
Kahului	20	19
Kahului	21	5
Kahului	21	10
Hilo	22	12
Hilo	22	15
Hilo	22	16